

AutoSmart Parts

Project Compliance Report

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This document verifies that the AutoSmart Parts project meets **ALL** requirements for both Phase 1 and Phase 2 of the AI Agent implementation.

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Phase 2: Order/Shipping Status & Fraudulent Transactions AI Agents

1. New AI Agents Created

Order/Shipping Status AI Agent

- **Implementation:** server.js lines 1146-1213
- **Function:** `analyzeImageWithAI(imageBase64, issueType, orderDetails)`
- **Use Cases:**
 - Defective product detection
 - Damaged shipping box detection
- **Technology:** GPT-4o-mini with vision capability
- **Integration:** Backend routing via support ticket system

Fraudulent Transactions AI Agent

- **Implementation:** Same `analyzeImageWithAI` function with `issueType: 'fraud'`
- **Capability:** OCR text extraction from credit card statements/receipts
- **Technology:** GPT-4o-mini vision + text analysis
- **Routing:** server.js lines 1427-1428

App Server Routing Updated

- **Endpoint:** `POST /api/support/ticket` in server.js lines 1301-1448
- **Routing Logic:**
 - `if (issueType === 'fraud') → fraud_detection agent`
 - `if (issueType === 'damaged') → damage_assessment agent`
 - `if (issueType === 'defect') → defect_detection agent`
- **Decision Application:** `applyAiDecision()` function in server.js lines 1215-1299

2. Order/Shipping Status AI Agent - Use Cases

Use Case 1: Defective Product

Customer Workflow:

1. Customer messages about defective product
 - o Location: CustomerService.jsx
 - o Form Fields: Order selection dropdown, issue type "defect", description
2. Customer uploads picture of defective product
 - o Implementation: File upload with preview
 - o Backend: Multer middleware handles file upload
3. AI Agent analyzes picture using Image Analysis model
 - o Model: GPT-4o-mini with vision capability
 - o Function: `analyzeImageWithAI()`
4. Agent makes decision: Refund, Replace, or Escalate
 - o REFUND: Marks ticket resolved, attempts order cancellation
 - o REPLACE: Marks ticket in_progress for replacement processing
 - o ESCALATE: Escalates to human agent

Use Case 2: Damaged Shipping Box

Customer Workflow:

1. Customer messages about damaged shipping box
 - o Form: Same support ticket form, issue type "damaged"
2. Customer uploads picture of damaged box
 - o Frontend: Image preview before submission
3. AI Agent analyzes damage using image analysis
 - o Same Model: GPT-4o-mini vision
 - o Context: Order details provided to AI for verification
4. Agent makes decision: Refund, Replace, or Escalate
 - o REFUND: If box damage likely caused product damage
 - o REPLACE: If box damaged but product likely intact
 - o ESCALATE: If uncertain or needs human review

3. Fraudulent Transactions AI Agent - Use Case

Use Case: Fraudulent Charge Report

Customer Workflow:

1. Customer messages to report fraudulent transaction
 - Frontend: CustomerService.jsx
 - Issue Type: "fraud"
2. Customer uploads picture of credit card statement/receipt (OCR)
 - File Upload: Supports JPG, PNG, PDF
 - Preview: Shows uploaded image before submission
3. AI Agent uses OCR/Text Extraction to read text from image
 - Model: GPT-4o-mini vision with OCR capability
 - Extraction: AI extracts text from statement image
4. Agent uses Text Analysis/LLM to understand context
 - Analysis: Compares extracted transaction details against order records
 - Verification: Checks transaction amount, date, merchant name
5. Agent makes decision: Refund, Decline, or Escalate
 - REFUND: If transaction appears fraudulent
 - DECLINE: If transaction matches legitimate order

Technical Implementation Details

Database Schema

1. User Management

- **Users** - Customer accounts and authentication
- **Addresses** - Shipping/billing addresses

2. Product Catalog

- **Products** - Product information and pricing
- **Categories** - Product categorization
- **Product Images** - Product photos and media

3. Order Management

- **Orders** - Customer orders and payments
- **Order Items** - Individual order line items
- **Shopping Carts** - Active shopping sessions
- **Cart Items** - Products in carts

4. Customer Service

- **Support Tickets** - Customer support requests
- **Ticket Attachments** - Support ticket files/images
- **Ticket Messages** - Support conversations

5. AI & Analytics

- **AI Interactions** - AI agent logs and performance
- **Analytics Events** - User behavior tracking
- **Audit Logs** - System change history

6. Business Operations

- **Inventory Transactions** - Stock level changes
- **Product Reviews** - Customer ratings and reviews
- **Notifications** - User alerts and messages

AI & Analytics Links

AI Interactions → Users, Orders, Support Tickets

Analytics Events → Users, Products, Orders

Key Connections

- Shopping Carts → Cart Items → Products
- Products → Categories
- Orders → Order Items → Products

API Endpoints Summary

Endpoint	Method	Purpose	AI Agent
/chat	POST	Product recommendations	Product Recommendation
/api/support/ticket	POST	Create support ticket with AI analysis	Order/Shipping Status & Fraud Detection
/api/support/ticket/:id/reprocess	POST	Rerun AI analysis on existing ticket	All agents
/api/orders	POST	Complete purchase	-
/api/orders/my	GET	View order history	-
/analytics/report	GET	AI performance metrics	-

Conclusion

The AutoSmart Parts project is **FULLY COMPLIANT** with all Phase 1 and Phase 2 requirements.

All AI agents are implemented, integrated, and functional:

1. **Product Recommendation Agent** - Provides personalized product suggestions with Accept/Decline options
2. **Order/Shipping Status Agent** - Analyzes defect/damage images and makes decisions
3. **Fraudulent Transactions Agent** - Performs OCR and fraud detection

The system includes:

- Complete frontend UI with React
- Backend with Express and PostgreSQL
- Three distinct AI agents using OpenAI GPT-4o-mini
- Full purchase workflow with form validation
- Support ticket system with image analysis
- Analytics tracking for all interactions
- Enhanced login/signup pages
- Order status information in chatbot